

MEMORY

Headings

- Definition
- Stages of Memory
- Types of Memory
- Short-term Memory
- Long-term Memory
- Explicit / Declarative Memory
- Implicit / Non-declarative Memory
- Important Examples

1. Definition

- **Memory** = retention of learned information and experiences with the ability to **retrieve and utilize** them whenever required.

2. Stages of Memory

Easy sequence:

Register → Retain → Recall → Reutilize

1. Registration

- Proper **perception and attention** are required.
- Failure of perception/attention → failure of registration → impaired learning.

2. Integration & Retention

- Learned experiences are **processed and integrated in CNS**.
- Stored as **short-term or long-term memory**.

3. Recognition & Recall

- Stored information is **retrieved when required**.

4. Reutilization

- Retrieved memory is **utilized for further learning and performance**.

3. Types of Memory

Broad classification:

MEMORY

→ **Explicit / Declarative**

→ **Implicit / Non-declarative**

A. Explicit / Declarative Memory

- Associated with **awareness**.
- Memory of:
 - **Events** → **Episodic memory**
 - **Facts** → **Semantic memory**
- Depends mainly on **hippocampus and medial temporal lobe**.

B. Implicit / Non-declarative Memory

- **Not associated with awareness**.
- Includes:
 - **Priming**
 - **Procedural memory** → skills & habits
 - **Associative learning**
 - Classical conditioning
 - Operant conditioning
 - **Non-associative learning**
 - Habituation
 - Sensitization

4. Short-term Memory

- Lasts for **seconds to hours**.
- Information is temporarily available for processing.
- **Working memory** is a form of short-term memory.
- Short-term memory can be converted into long-term memory by **consolidation**.
- Memory traces are relatively **susceptible to disruption** by trauma and drugs.

Important:

Working memory → Prefrontal cortex

5. Long-term Memory

- Stores information for **years or even lifelong**.
- Memory traces are relatively **resistant to disruption**.
- Includes both:
 - Explicit memory
 - Implicit memory

Important:

Short-term memory → Consolidation → Long-term memory

6. Important Forms of Implicit Memory

Priming

- **Facilitation of recognition** of a word/object due to prior exposure.
- Example: better recall of a word when its **first few letters** are given.

Procedural Memory

- Memory for **skills and habits**.
- Once acquired → becomes **unconscious and automatic**.

Non-associative Learning

- Learning about a **single stimulus**.

Two forms:

Habituation

- Repeated neutral stimulus → **decreased response**.

Sensitization

- Repeated/strong stimulus → **increased response**.

Associative Learning

- Learning the **relationship between two stimuli/events**.
- Includes:
 - Classical conditioning
 - Operant conditioning

7. Important Brain Areas

Memory function	Main area
Working memory	Prefrontal cortex
Declarative memory / consolidation	Hippocampus + medial temporal lobe
Long-term storage	Various areas of neocortex
Procedural memory	Striatum + motor cortex + cerebellum

One-minute revision

MEMORY

Definition

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4 stages

→ Register → Retain → Recall → Reutilize

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2 major types

→ Explicit / Declarative

→ Implicit / Non-declarative

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Explicit

→ Episodic + Semantic

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Implicit

→ Priming + Procedural + Associative + Non-associative

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Duration

→ Short-term → Long-term

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Consolidation

→ STM → LTM